ASAF BELILUS

Software Engineer · M.Sc. Computer Science · Full-Stack & Computer Vision

CONTACT

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LinkedIn
GitHub
Portfolio

SKILLS & TECHNOLOGIES

Languages
Python, Java, TypeScript, C/C++

AI & Research
Computer vision pipelines, evaluation & regression testing, NumPy, SciPy

Backend & Data
Spring Boot, REST APIs, JWT auth, PostgreSQL

Tools & Testing
Docker, Git, automated test suites

EDUCATION

M.Sc. Computer Science
Ben-Gurion University of the Negev
Researcher, Prof. Gera Weiss's Lab · 2026 – Present (Expected Oct 2027)
Current GPA 95.

B.Sc. Computer Science
Ben-Gurion University of the Negev · 2022 – 2025
Final GPA 84.2 · Active swimmer, BGU ASA competitive swim team.
Coursework: Computational Models, Probability & Statistics, Intro to Computational Learning, Algorithms

ACTIVITIES

Teaching Assistant — BGU · 2026 – Present
Supervises practicum coursework; grades and examines student work.

Individual/Group Swimming Instructor — Wingate Institute · 2014 – Present
Teaches fundamentals and advanced technique to swimmers of all ages.

Volunteer Math & CS Tutor — BGU · 2026 – Present
Tutors first-year students in math and CS, focused on reservist students returning to their studies.

LANGUAGES

Hebrew — Native
English — Fluent

PROFILE

Computer Science M.Sc. student seeking a software engineering student position. Combines end-to-end development on SwimEdge — Spring Boot, React, and PostgreSQL — with Python research pipelines for underwater motion analysis. Works across disciplines and with external stakeholders, turning ambiguous requirements into tested software.

RESEARCH

Swimmer Biomechanics Research — Prof. Gera Weiss's Lab

Ben-Gurion University of the Negev · in collaboration with Prof. Raziel Reimer, Faculty of Engineering · 2026 – Present

- Conducting cross-disciplinary research applying computer vision and control theory to competitive-swimming biomechanics, bridging the Computer Science and Engineering faculties.
- Built Python pipelines that reconstruct swimmer joint motion from 3-D underwater pose-estimation data, using forward/inverse kinematics and nonlinear least-squares optimization to drive a third-party hydrodynamic simulation engine.
- Diagnosed systematic model-fitting errors (degenerate optimization bounds, skeletal geometry mismatches) via quantitative probing against ground-truth motion-capture data, and built the evaluation and golden-file regression suite that separates code error from data limits.

FEATURED PROJECT

SwimEdge — Full-Stack Competitive Swimming Platform

Founder & Sole Engineer · 2025 – Present · GitHub

- Took a full-stack platform for swimmer analytics and competition management from concept to a working system as the sole engineer, motivated by a gap identified firsthand as a competitive swimmer.
- Engineered a modular, service-oriented Spring Boot + PostgreSQL backend and a custom PDF/Excel parsing pipeline for Israeli Swimming Association data, backed by automated test suites and Docker container builds.
- Engaged external stakeholders directly — the Israeli Swimming Association and the Ministry of Sport — translating federation regulations into system requirements and presenting the platform to leadership.

MILITARY SERVICE

Combat Commander, Paratrooper Brigade, IDF

2016 – 2019

- Trained and mentored newly recruited soldiers; led operations and managed company-level logistics.

Reservist, Paratrooper Brigade, IDF

2019 – Present

- Lead frontline teams under pressure, maintaining calm and accountability.